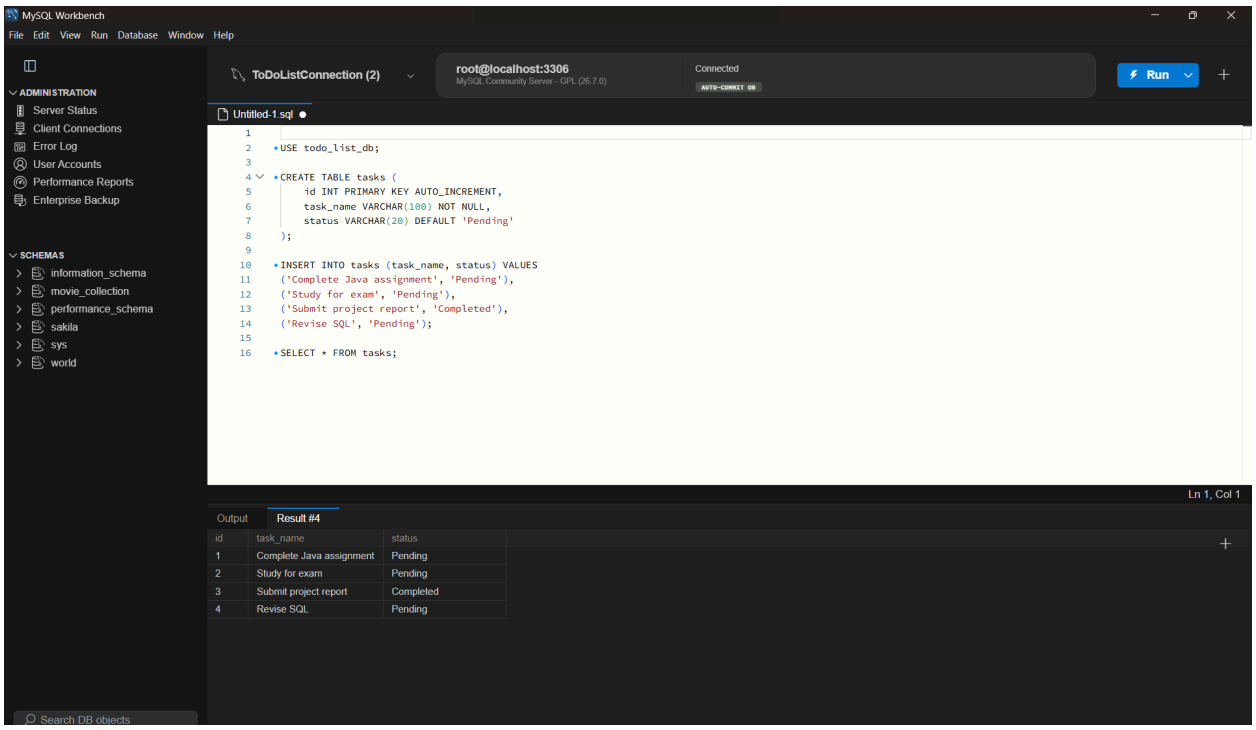
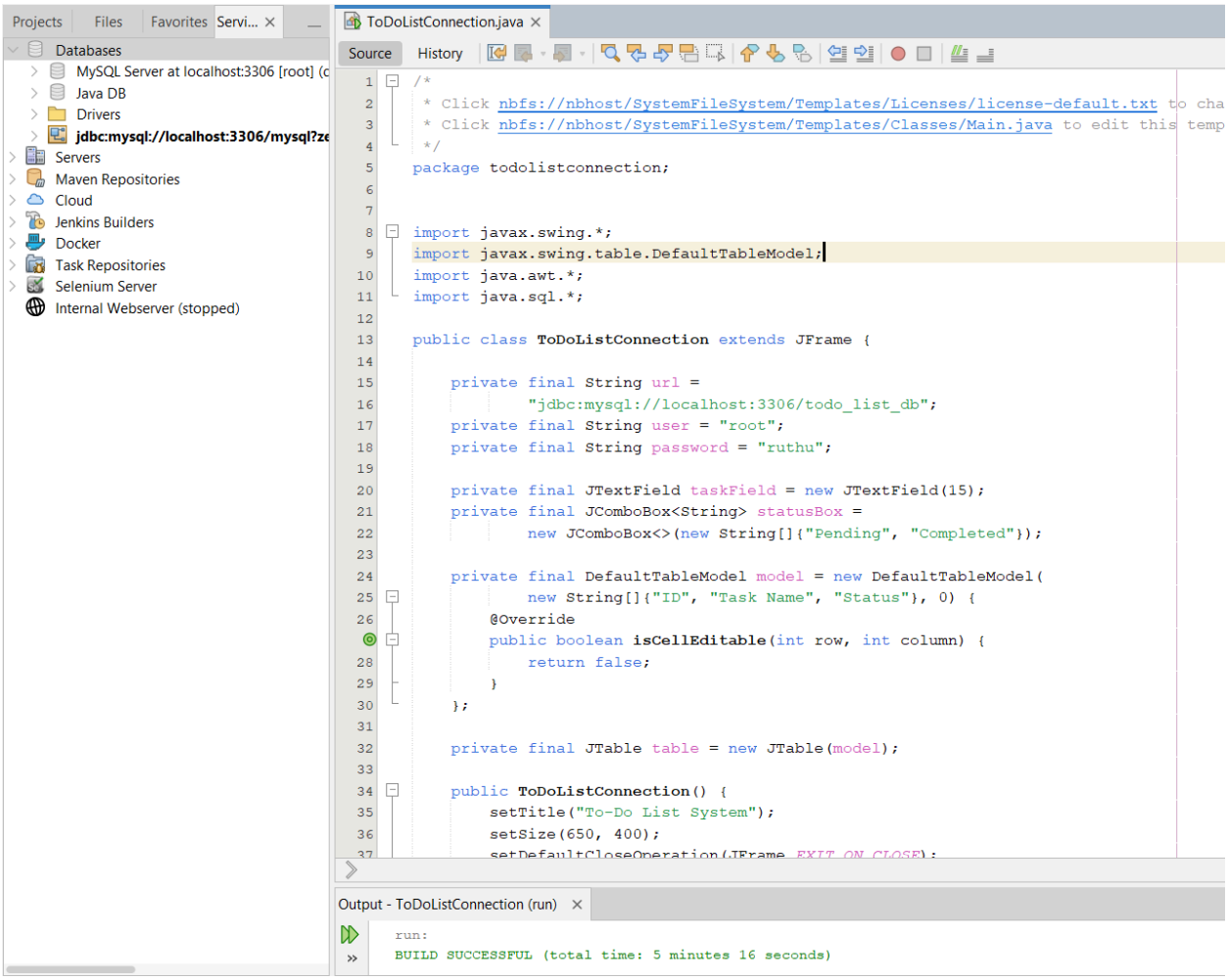
Mini Project

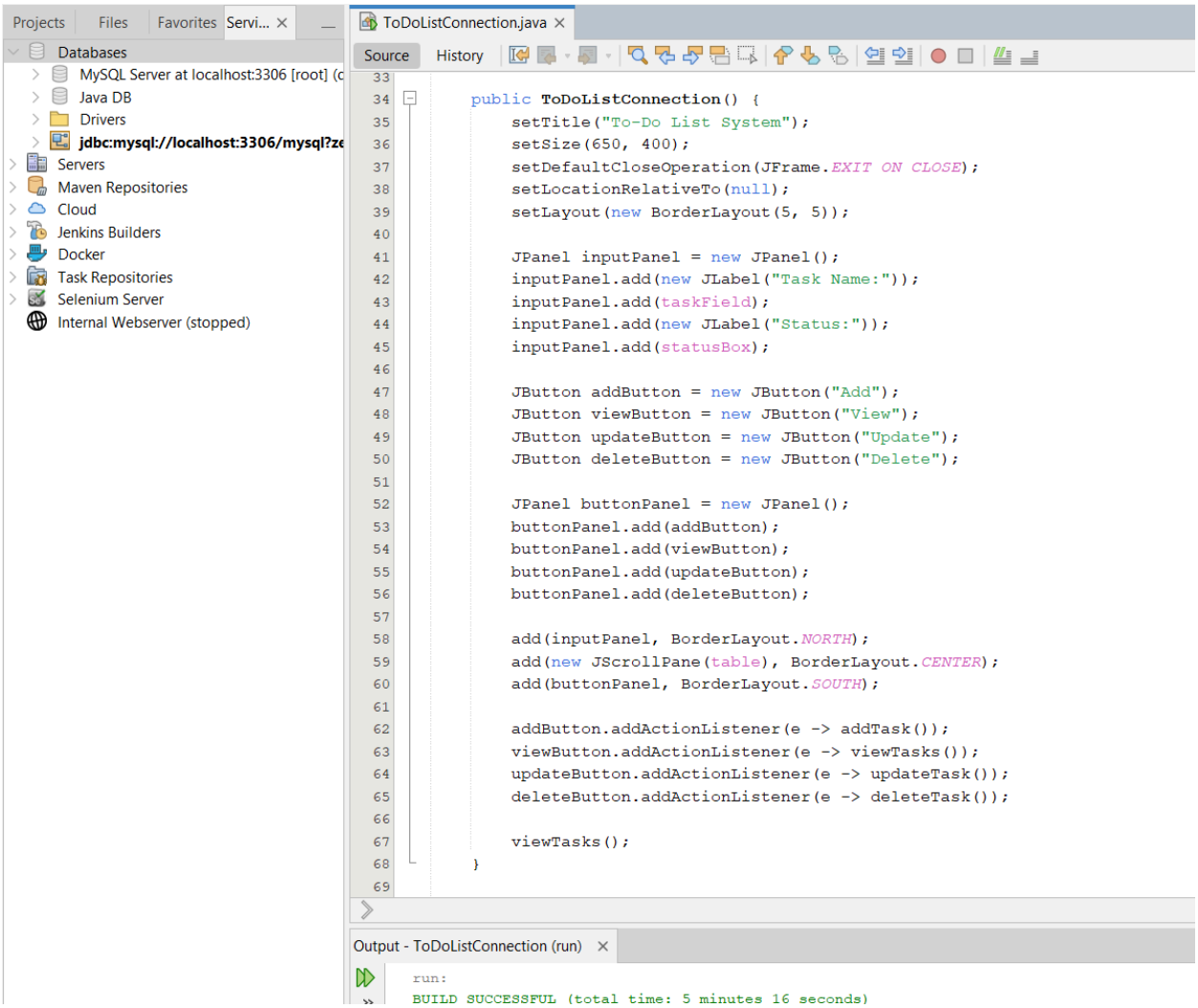
Topic: TO DO LIST

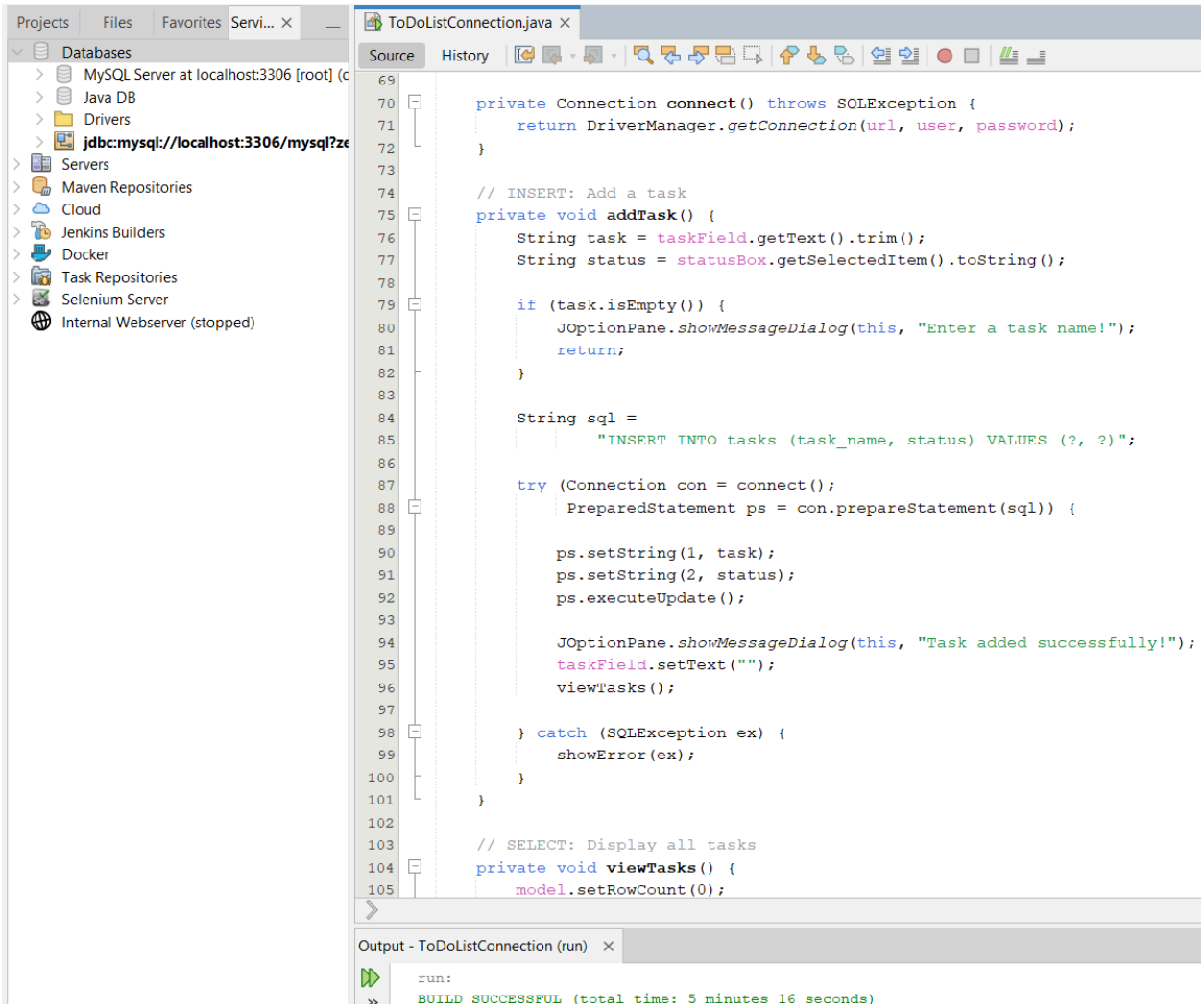
Mysql Workbench Database:

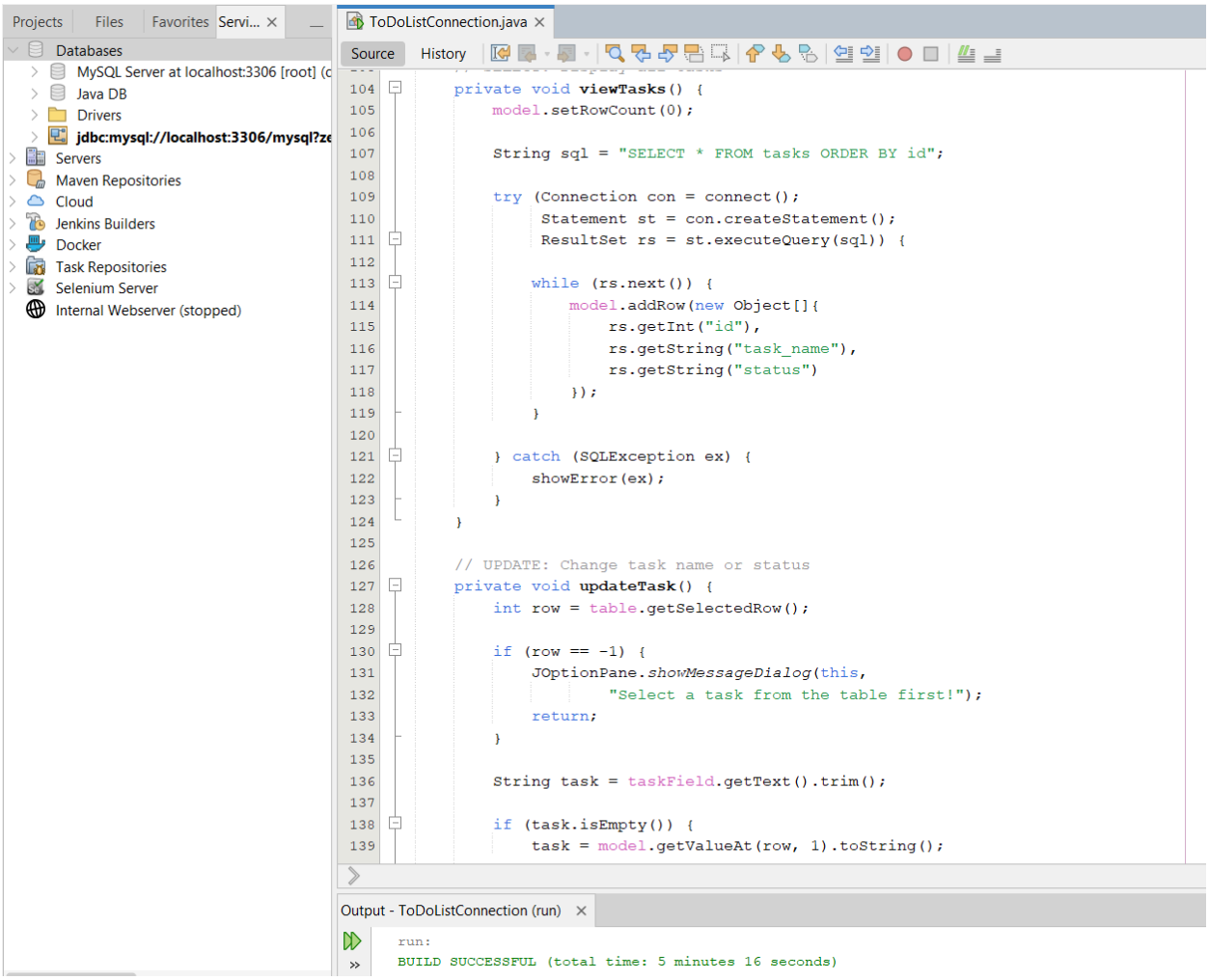


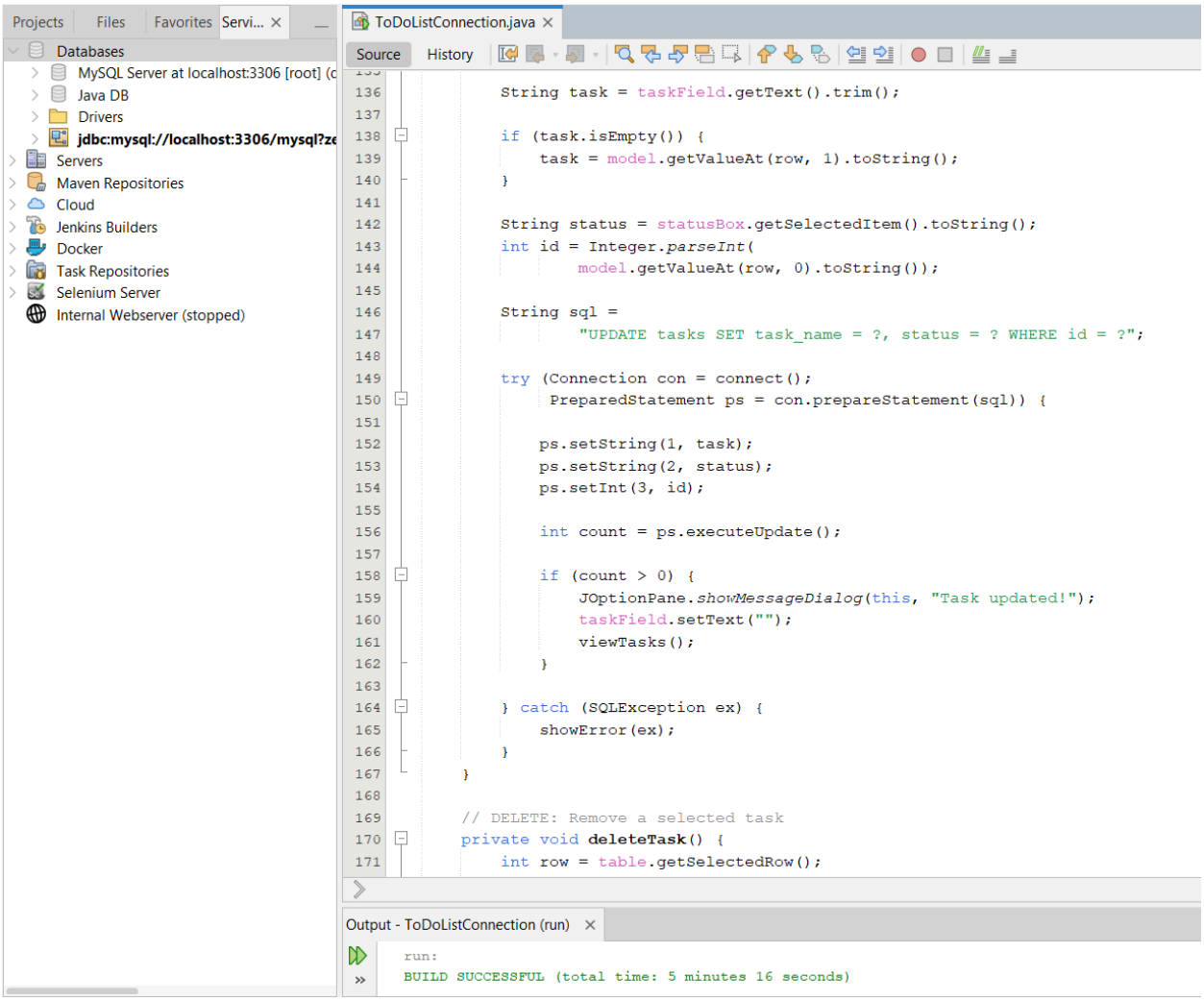
Java Application:

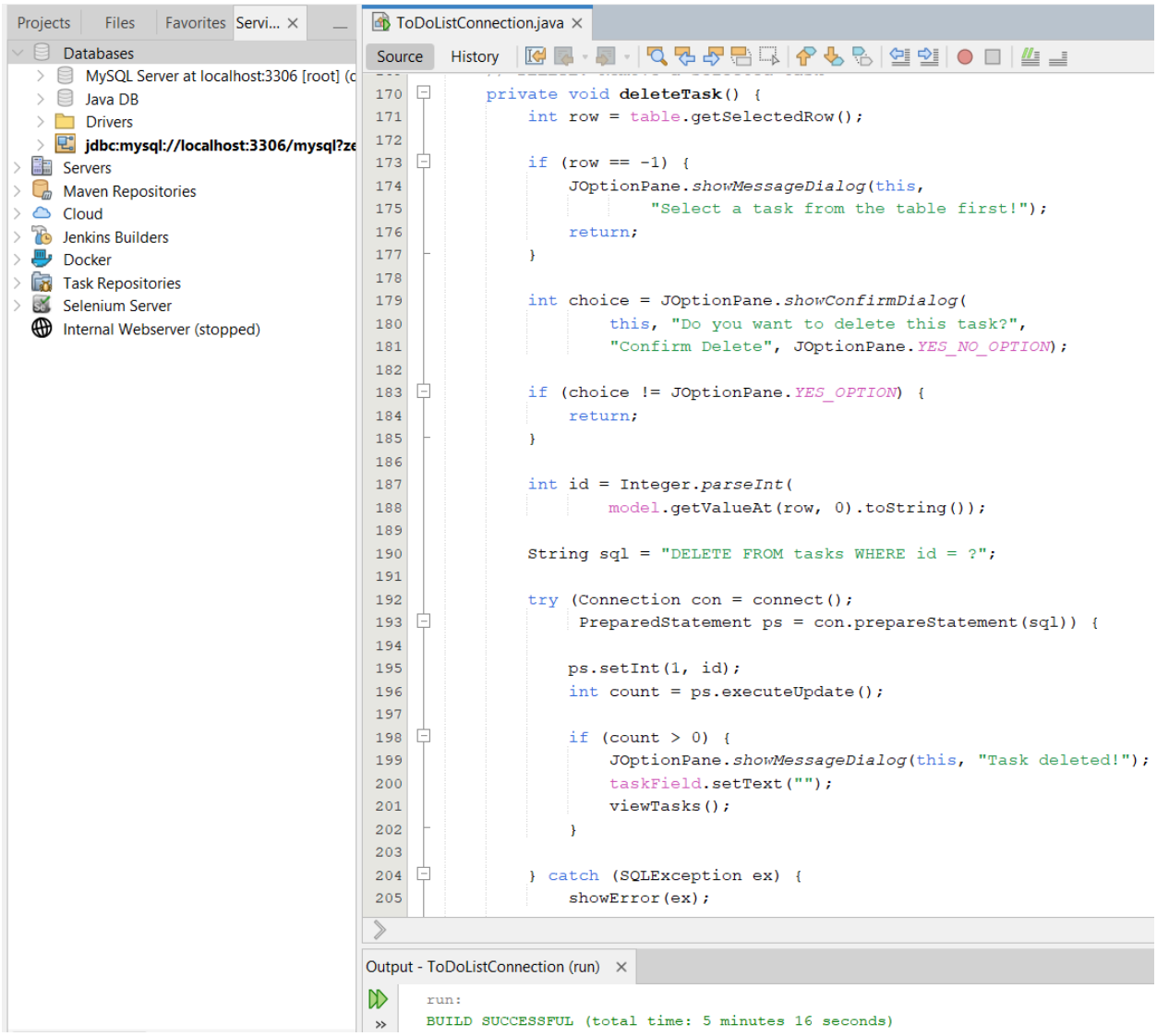


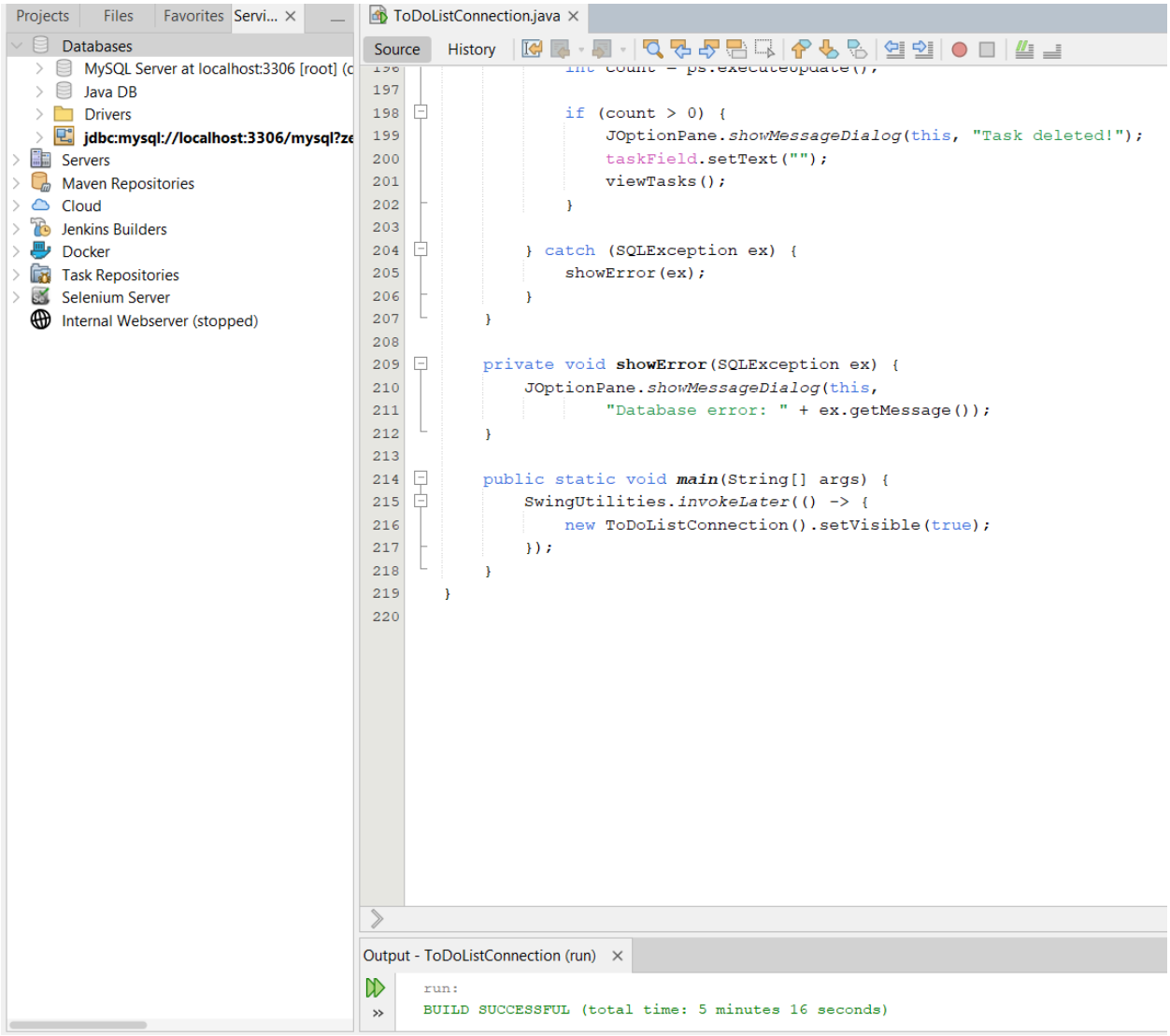

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Selenium Server
Internal Webserver (stopped)
ToDoListConnection.java
Source
History
33
34 public ToDoListConnection() {
35 setTitle("To-Do List System");
36 setSize(650, 400);
37 setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
38 setLocationRelativeTo(null);
39 setLayout(new BorderLayout(5, 5));
40
41 JPanel inputPanel = new JPanel();
42 inputPanel.add(new JLabel("Task Name:"));
43 inputPanel.add(taskField);
44 inputPanel.add(new JLabel("Status:"));
45 inputPanel.add(statusBox);
46
47 JButton addButton = new JButton("Add");
48 JButton viewButton = new JButton("View");
49 JButton updateButton = new JButton("Update");
50 JButton deleteButton = new JButton("Delete");
51
52 JPanel buttonPanel = new JPanel();
53 buttonPanel.add(addButton);
54 buttonPanel.add(viewButton);
55 buttonPanel.add(updateButton);
56 buttonPanel.add(deleteButton);
57
58 add(inputPanel, BorderLayout.NORTH);
59 add(new JScrollPane(table), BorderLayout.CENTER);
60 add(buttonPanel, BorderLayout.SOUTH);
61
62 addButton.addActionListener(e -> addTask());
63 viewButton.addActionListener(e -> viewTasks());
64 updateButton.addActionListener(e -> updateTask());
65 deleteButton.addActionListener(e -> deleteTask());
66
67 viewTasks();
68 }
69
Output - ToDoListConnection (run)
run:
BUILD SUCCESSFUL (total time: 5 minutes 16 seconds)


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ToDoListConnection.java
Source
History
69
70 private Connection connect() throws SQLException {
71 return DriverManager.getConnection(url, user, password);
72 }
73
74 // INSERT: Add a task
75 private void addTask() {
76 String task = taskField.getText().trim();
77 String status = statusBox.getSelectedItem().toString();
78
79 if (task.isEmpty()) {
80 JOptionPane.showMessageDialog(this, "Enter a task name!");
81 return;
82 }
83
84 String sql =
85 "INSERT INTO tasks (task_name, status) VALUES (?, ?)";
86
87 try (Connection con = connect();
88 PreparedStatement ps = con.prepareStatement(sql)) {
89
90 ps.setString(1, task);
91 ps.setString(2, status);
92 ps.executeUpdate();
93
94 JOptionPane.showMessageDialog(this, "Task added successfully!");
95 taskField.setText("");
96 viewTasks();
97
98 } catch (SQLException ex) {
99 showError(ex);
100 }
101 }
102
103 // SELECT: Display all tasks
104 private void viewTasks() {
105 model.setRowCount(0);
Output - ToDoListConnection (run)
run:
BUILD SUCCESSFUL (total time: 5 minutes 16 seconds)

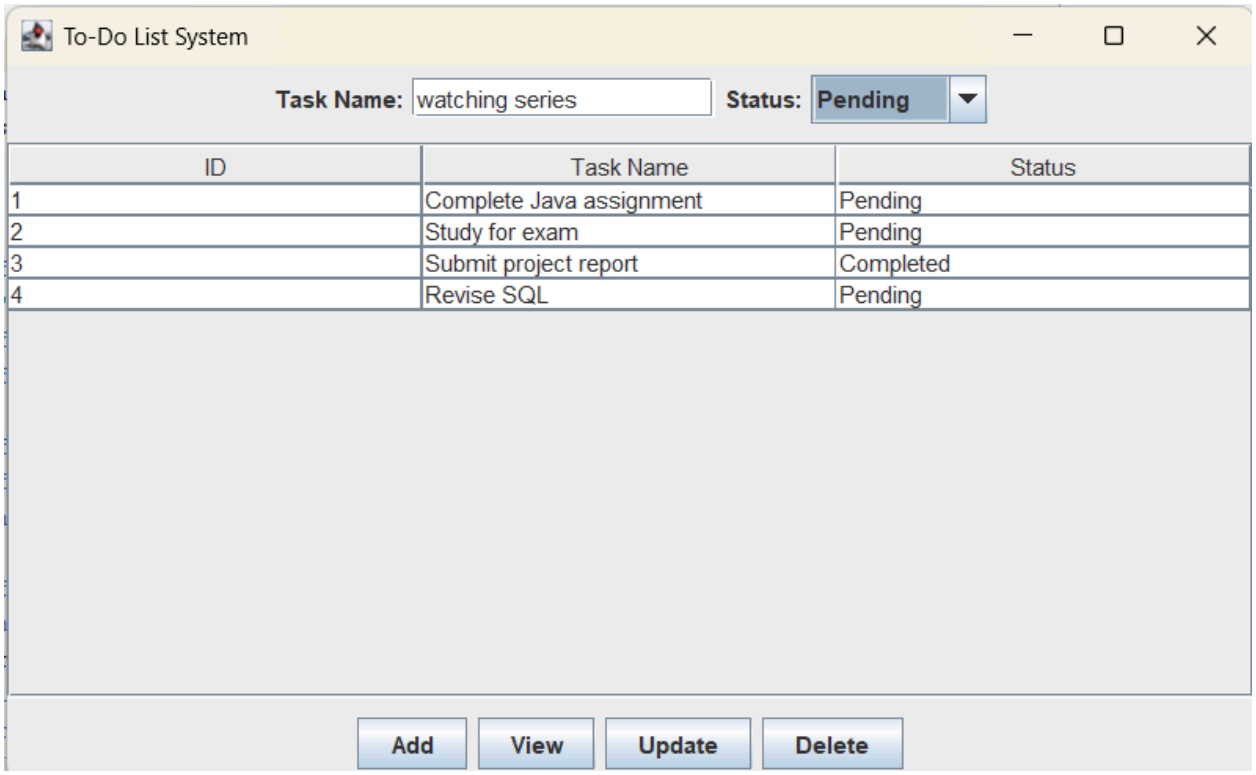

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ToDoListConnection.java
Source
History
104 private void viewTasks() {
105 model.setRowCount(0);
106
107 String sql = "SELECT * FROM tasks ORDER BY id";
108
109 try (Connection con = connect();
110 Statement st = con.createStatement();
111 ResultSet rs = st.executeQuery(sql)) {
112
113 while (rs.next()) {
114 model.addRow(new Object[]{
115 rs.getInt("id"),
116 rs.getString("task_name"),
117 rs.getString("status")
118 });
119 }
120
121 } catch (SQLException ex) {
122 showError(ex);
123 }
124 }
125
126 // UPDATE: Change task name or status
127 private void updateTask() {
128 int row = table.getSelectedRow();
129
130 if (row == -1) {
131 JOptionPane.showMessageDialog(this,
132 "Select a task from the table first!");
133 return;
134 }
135
136 String task = taskField.getText().trim();
137
138 if (task.isEmpty()) {
139 task = model.getValueAt(row, 1).toString();
Output - ToDoListConnection (run)
run:
BUILD SUCCESSFUL (total time: 5 minutes 16 seconds)

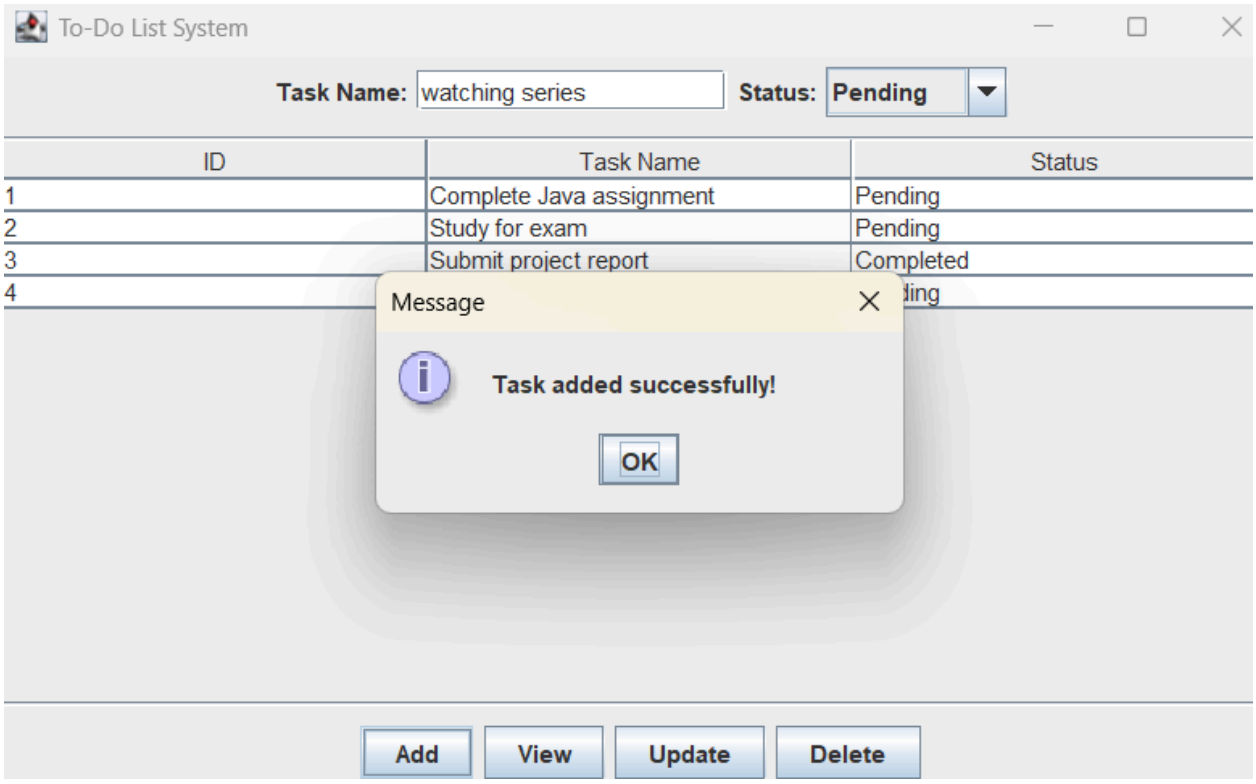

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ToDoListConnection.java
Source
History
String task = taskField.getText().trim();
if (task.isEmpty()) {
task = model.getValueAt(row, 1).toString();
}
String status = statusBox.getSelectedItem().toString();
int id = Integer.parseInt(
model.getValueAt(row, 0).toString());
String sql =
"UPDATE tasks SET task_name = ?, status = ? WHERE id = ?";
try (Connection con = connect();
PreparedStatement ps = con.prepareStatement(sql)) {
ps.setString(1, task);
ps.setString(2, status);
ps.setInt(3, id);
int count = ps.executeUpdate();
if (count > 0) {
JOptionPane.showMessageDialog(this, "Task updated!");
taskField.setText("");
viewTasks();
}
} catch (SQLException ex) {
showError(ex);
}
}
// DELETE: Remove a selected task
private void deleteTask() {
int row = table.getSelectedRow();
Output - ToDoListConnection (run)
run:
BUILD SUCCESSFUL (total time: 5 minutes 16 seconds)


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ToDoListConnection.java
Source
History
170 private void deleteTask() {
171 int row = table.getSelectedRow();
172
173 if (row == -1) {
174 JOptionPane.showMessageDialog(this,
175 "Select a task from the table first!");
176 return;
177 }
178
179 int choice = JOptionPane.showConfirmDialog(
180 this, "Do you want to delete this task?",
181 "Confirm Delete", JOptionPane.YES_NO_OPTION);
182
183 if (choice != JOptionPane.YES_OPTION) {
184 return;
185 }
186
187 int id = Integer.parseInt(
188 model.getValueAt(row, 0).toString());
189
190 String sql = "DELETE FROM tasks WHERE id = ?";
191
192 try (Connection con = connect();
193 PreparedStatement ps = con.prepareStatement(sql)) {
194
195 ps.setInt(1, id);
196 int count = ps.executeUpdate();
197
198 if (count > 0) {
199 JOptionPane.showMessageDialog(this, "Task deleted!");
200 taskField.setText("");
201 viewTasks();
202 }
203
204 } catch (SQLException ex) {
205 showError(ex);
Output - ToDoListConnection (run)
run:
BUILD SUCCESSFUL (total time: 5 minutes 16 seconds)

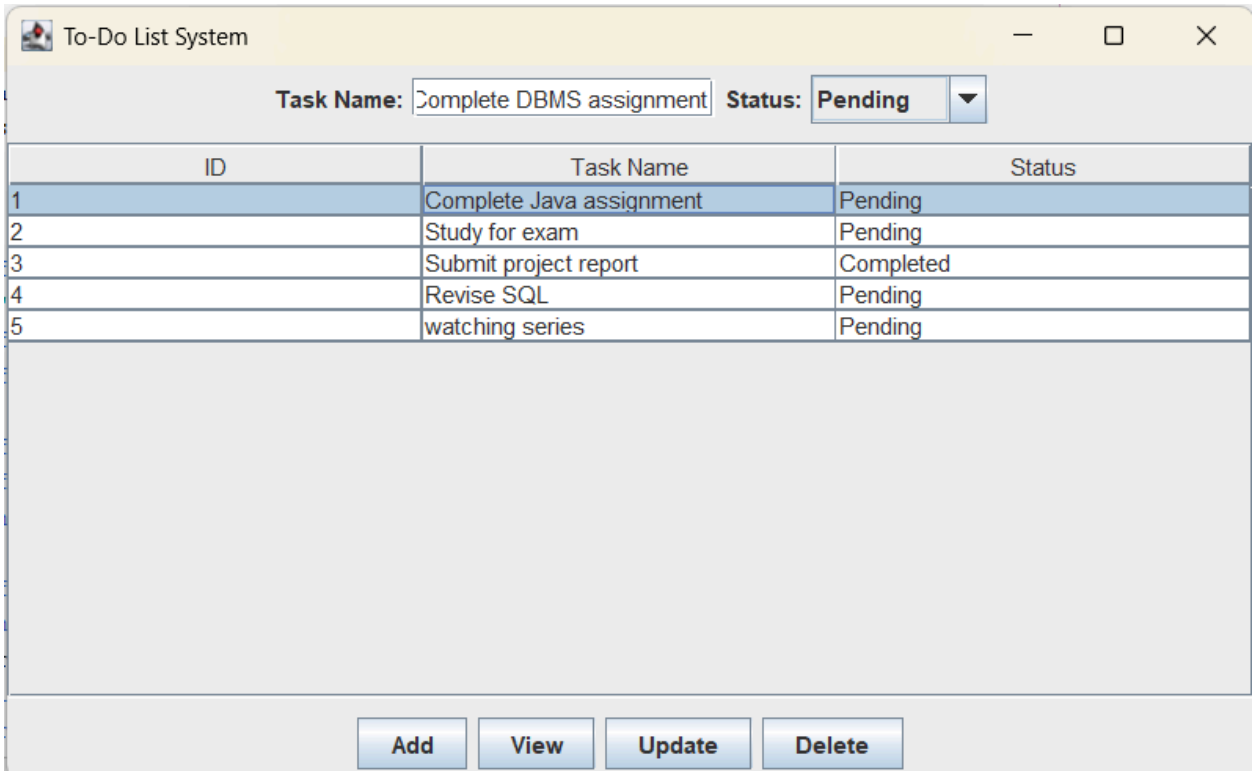

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ToDoListConnection.java
Source
History
int count = ps.executeUpdate();
if (count > 0) {
JOptionPane.showMessageDialog(this, "Task deleted!");
taskField.setText("");
viewTasks();
}
} catch (SQLException ex) {
showError(ex);
}
}
private void showError(SQLException ex) {
JOptionPane.showMessageDialog(this,
"Database error: " + ex.getMessage());
}
public static void main(String[] args) {
SwingUtilities.invokeLater(() -> {
new ToDoListConnection().setVisible(true);
});
}
}
Output - ToDoListConnection (run)
run:
BUILD SUCCESSFUL (total time: 5 minutes 16 seconds)

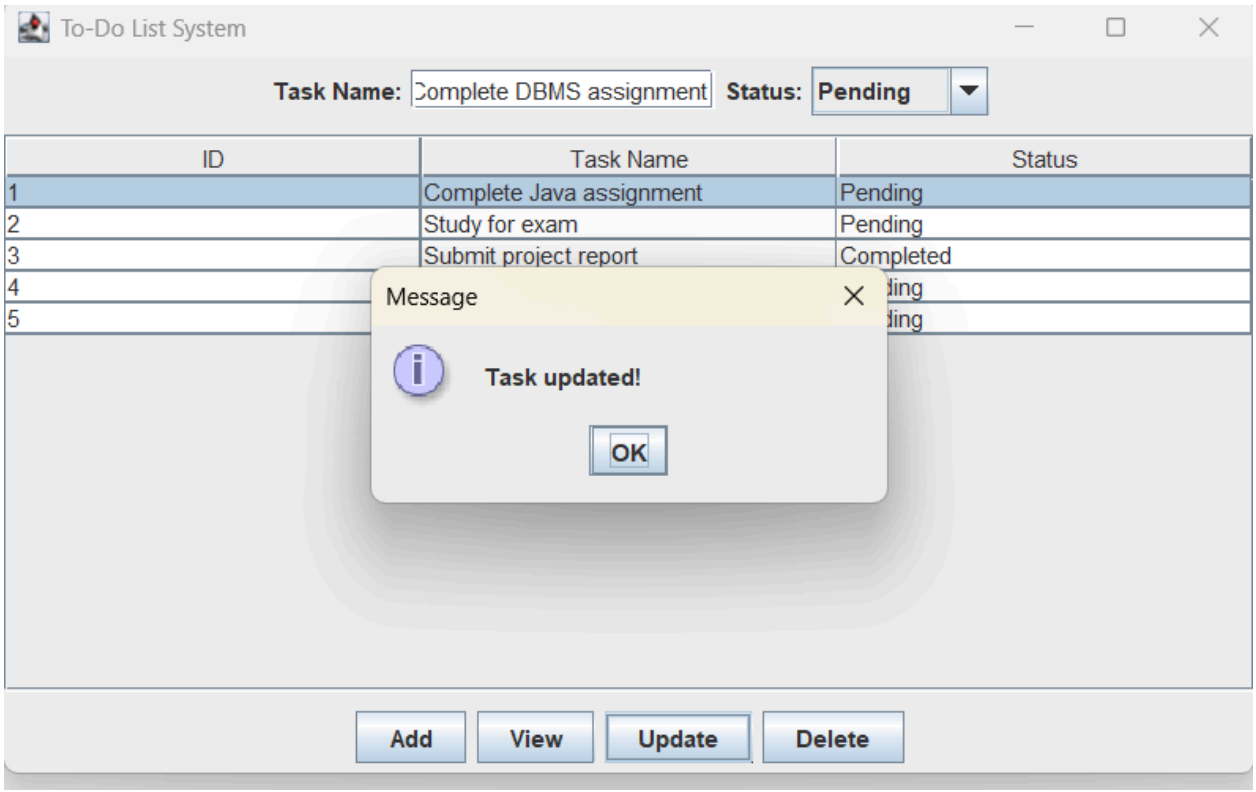
Adding an Task:

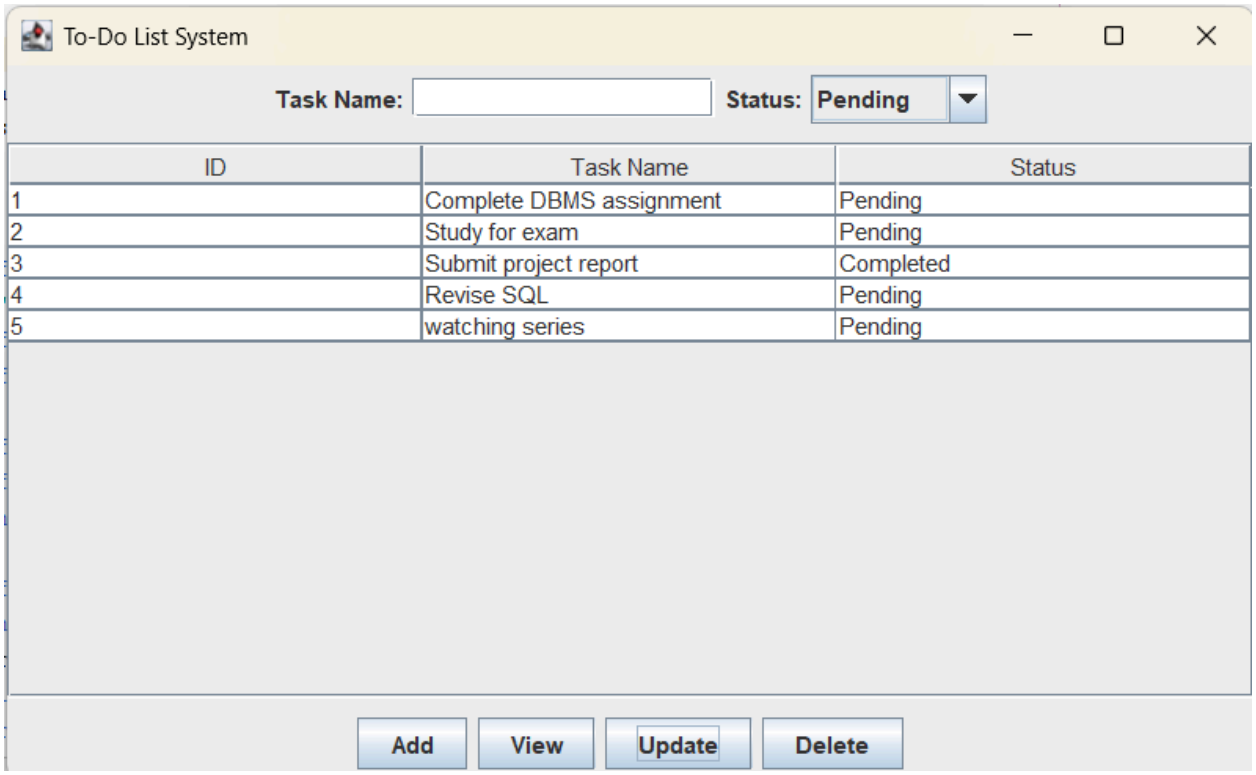




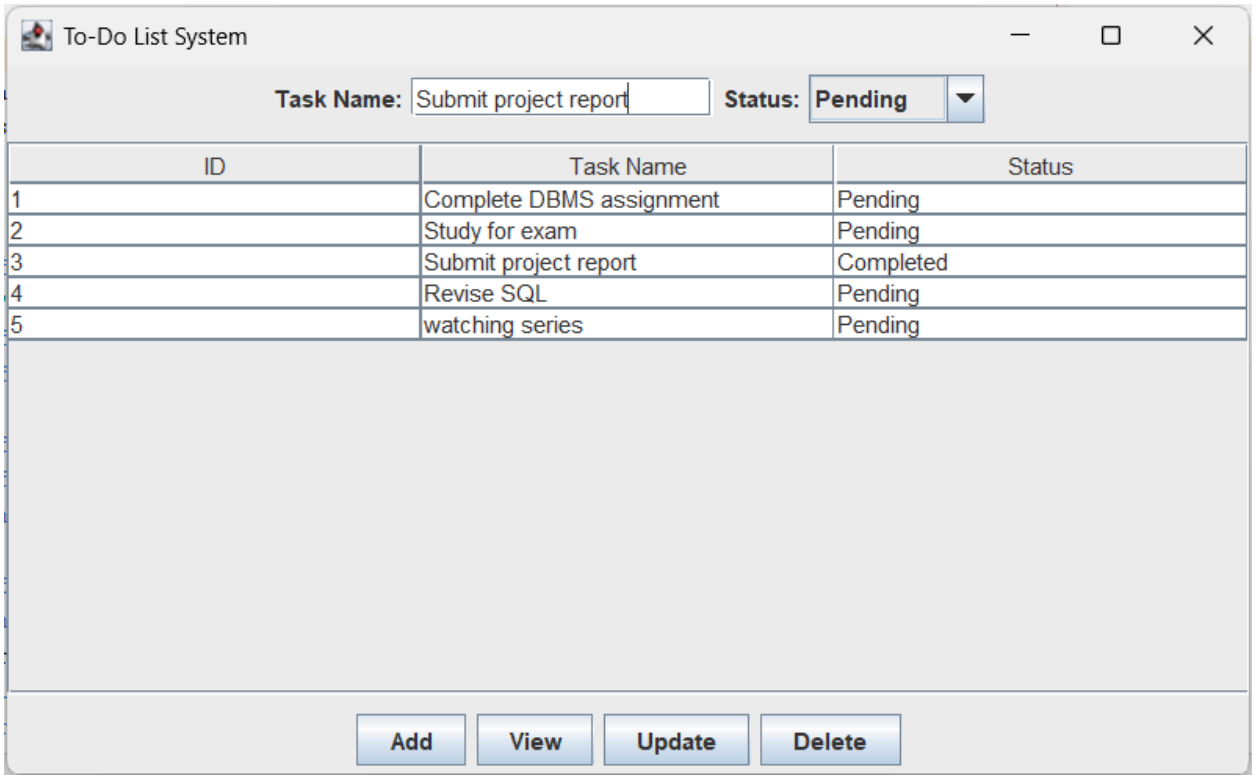
Updating a Task:

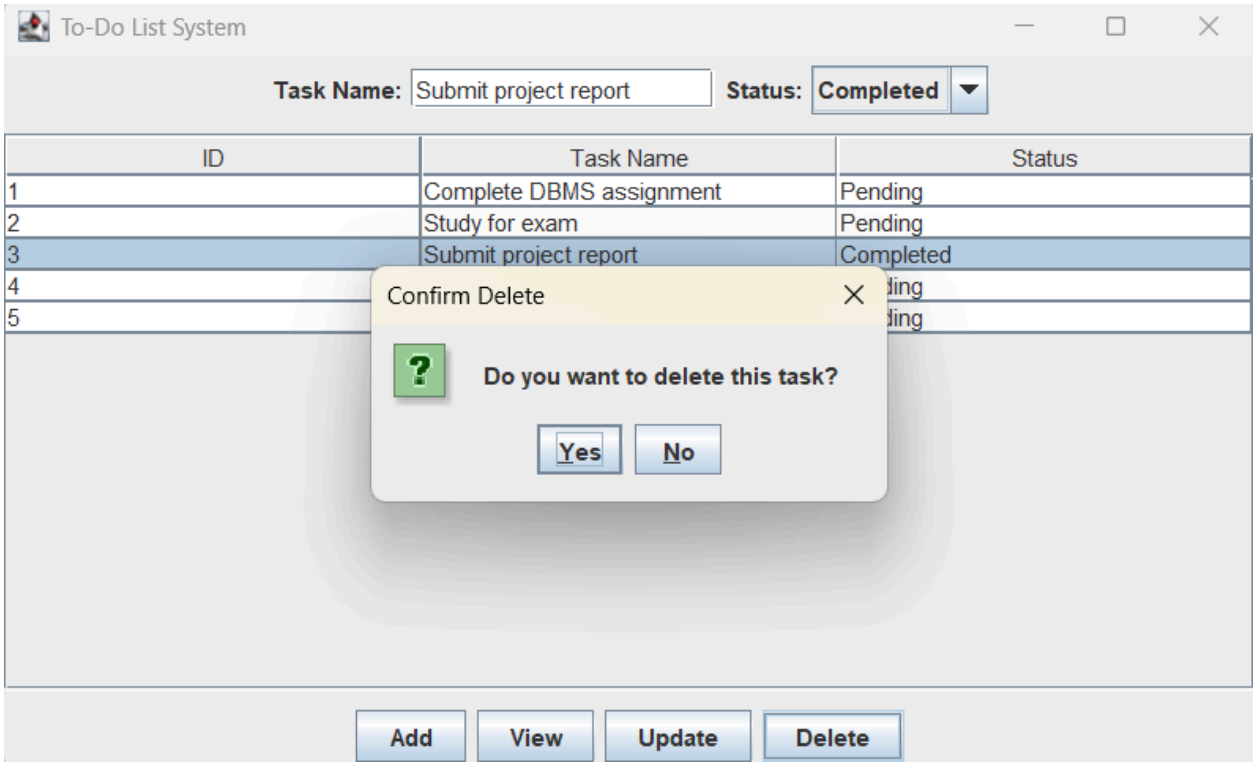


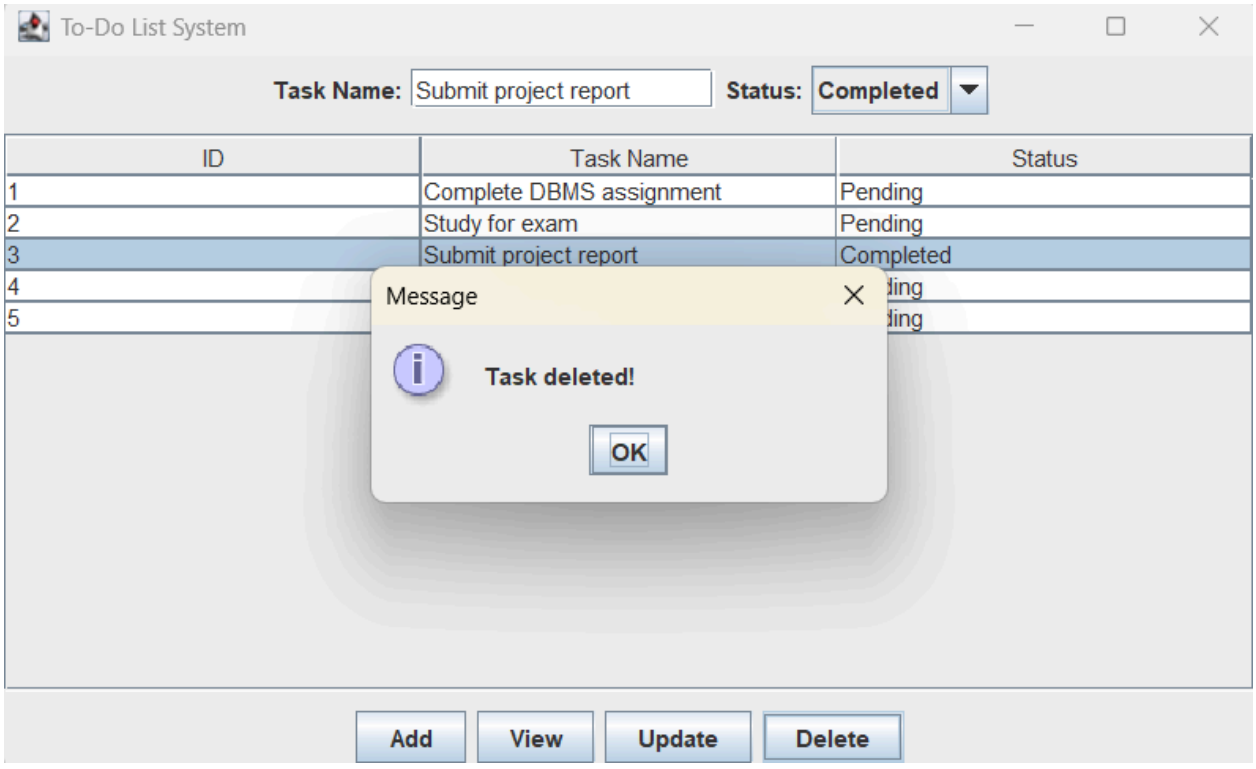


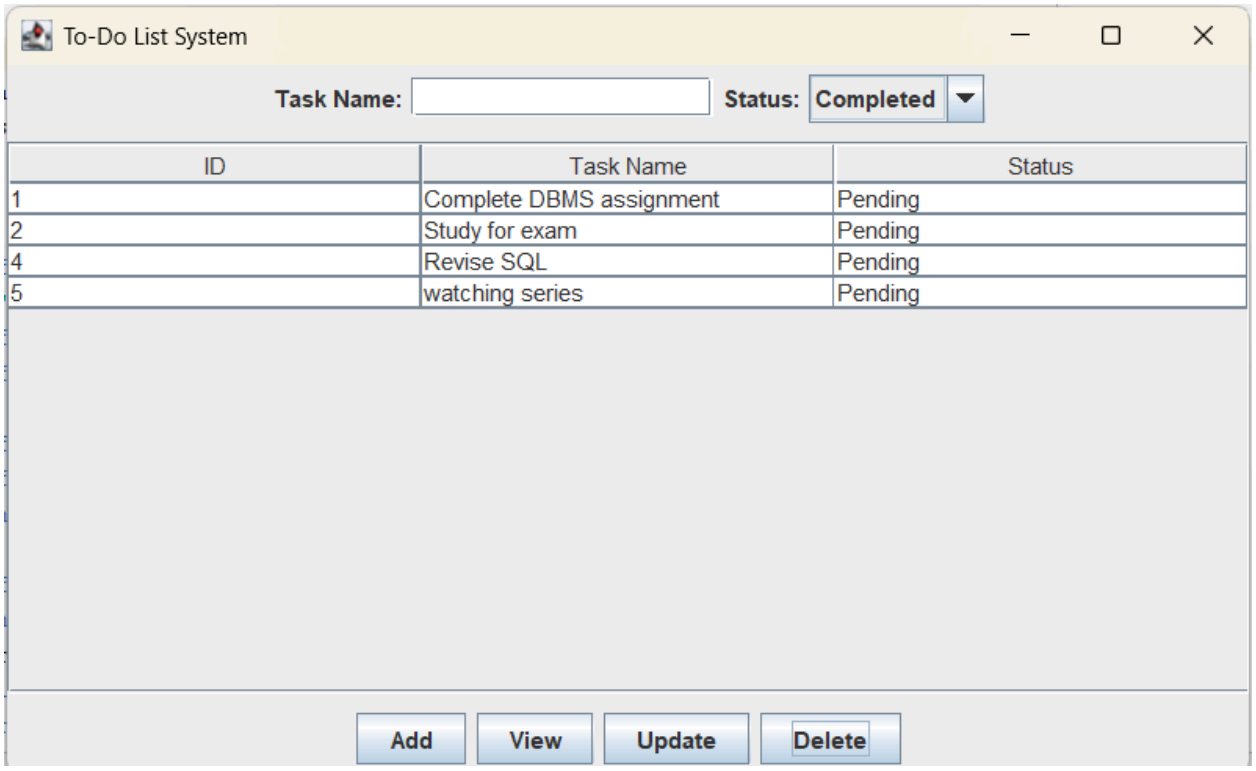


Deleting a Task:




To-Do List System
Task Name: Submit project report
Status: Completed
ID
Task Name
Status
1
Complete DBMS assignment
Pending
2
Study for exam
Pending
3
Submit project report
Completed
4
5
Confirm Delete
Do you want to delete this task?
Yes
No
Add
View
Update
Delete


To-Do List System
Task Name: Submit project report
Status: Completed
ID
Task Name
Status
1
Complete DBMS assignment
Pending
2
Study for exam
Pending
3
Submit project report
Completed
4
5
Message
Task deleted!
OK
Add
View
Update
Delete


To-Do List System
Task Name:
Status:
Completed
ID
Task Name
Status
1
Complete DBMS assignment
Pending
2
Study for exam
Pending
4
Revise SQL
Pending
5
watching series
Pending
Add
View
Update
Delete